Newton School of Technology

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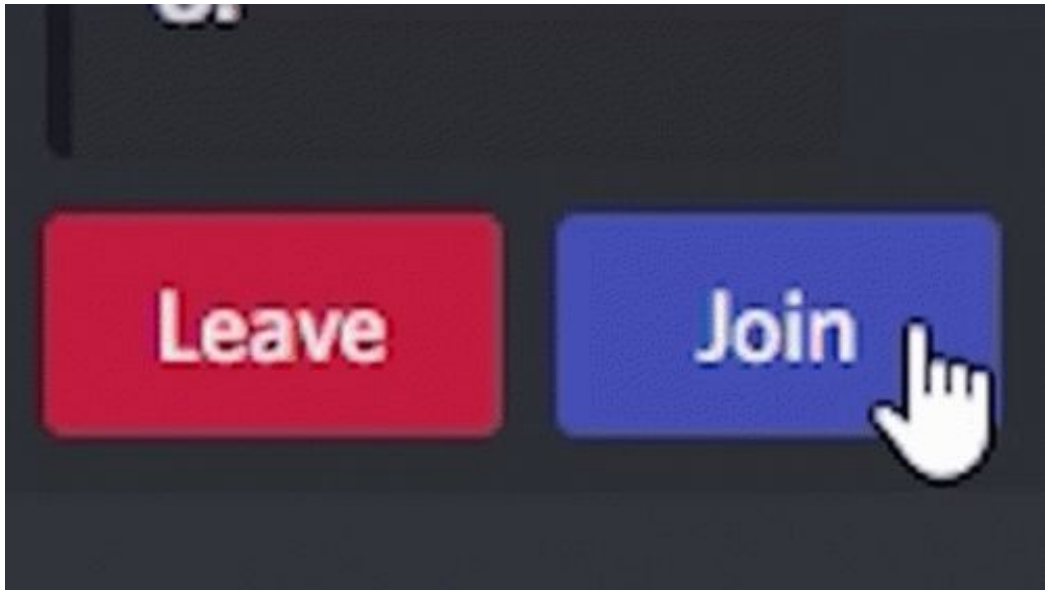
# Intro to AI-ML

# Join the Lecture on NS Portal :)



# Instructors

Shreyas Sir

Ankush Sir

Dev Sir

# Rules

1. No stupid questions policy. We do not tag any question as stupid, feel free to ask anything your ❤ desires in class or otherwise.
2. We never say no to questions, but we might reschedule it or give you the resources to find answers
3. Our purpose is to train you all to become good engineers and exceptional humans.
4. We take academic integrity and discipline very very seriously.
5. We do not let our personal life impact you; if we scold you it is ~~mostly~~ because of something you did.

* This is not an exhaustive list of rules…

# Rules

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1. 75% attendance mandatory as per university rules
2. No late entries permitted
3. Laptop screens down is the default state unless specified otherwise
4. Using phones, smartwatches, earphones or any other electronic devices is NOT permitted in class unless instructed otherwise.
5. Students must occupy the front seats first, class would not start if the front seats are vacant
6. When you have a doubt in class, raise your hand, stand up when pointed by the instructor and then ask!
7. Be respectful to everyone.

* This is not an exhaustive list of rules…

# What are the necessities of life?

# Internet

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Gemini

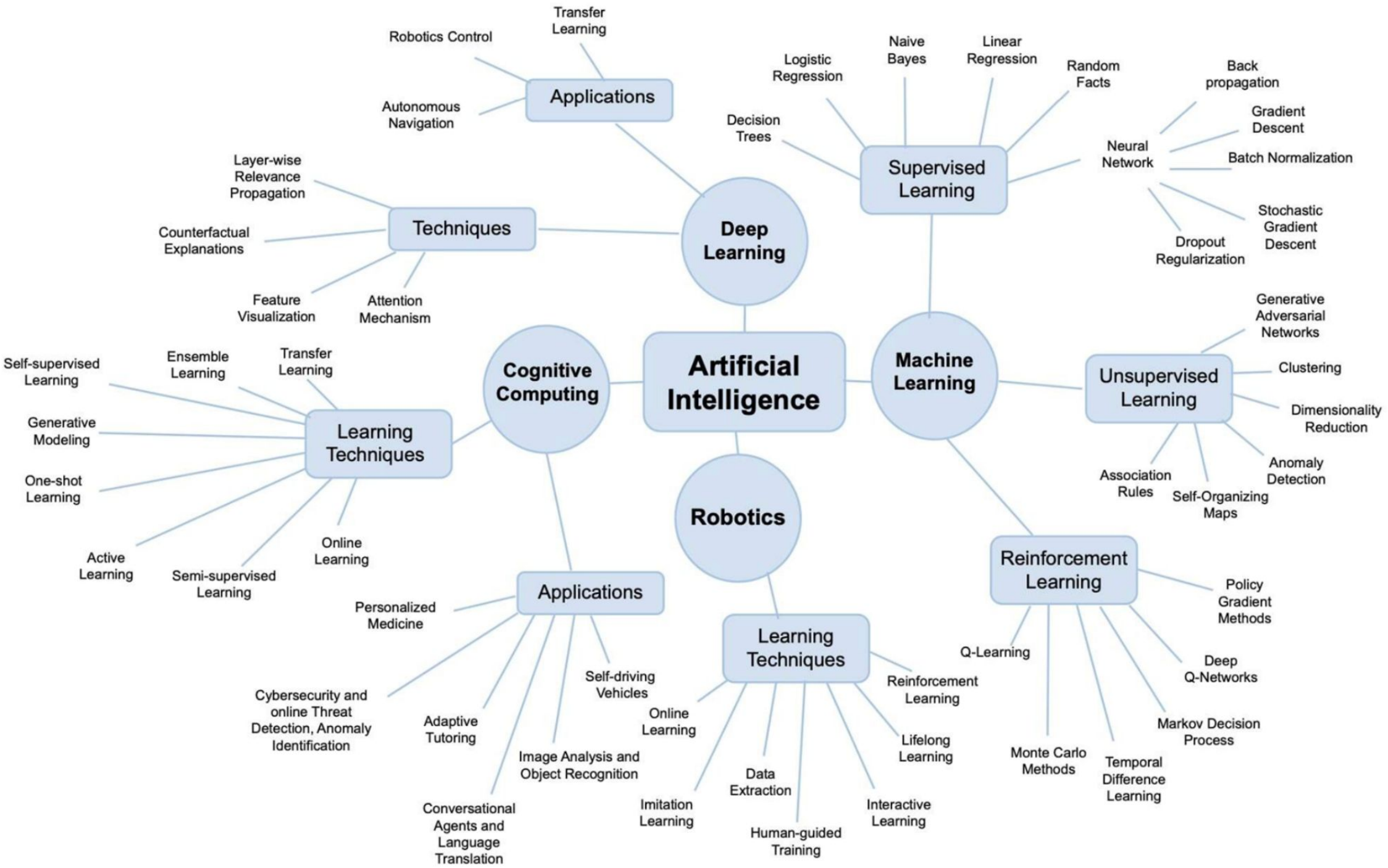
replit

**How many of you have used this?**

# Problems with AI

- Hallucinations
- Same model gives different output? Why?

# The Present and Future


Artificial Intelligence
Deep Learning
Applications
Transfer Learning
Robotics Control
Autonomous Navigation
Techniques
Layer-wise Relevance Propagation
Counterfactual Explanations
Feature Visualization
Attention Mechanism
Machine Learning
Supervised Learning
Logistic Regression
Naive Bayes
Linear Regression
Random Facts
Decision Trees
Neural Network
Back propagation
Gradient Descent
Batch Normalization
Stochastic Gradient Descent
Dropout Regularization
Unsupervised Learning
Generative Adversarial Networks
Clustering
Dimensionality Reduction
Anomaly Detection
Self-Organizing Maps
Association Rules
Reinforcement Learning
Policy Gradient Methods
Deep Q-Networks
Markov Decision Process
Temporal Difference Learning
Monte Carlo Methods
Q-Learning
Cognitive Computing
Learning Techniques
Self-supervised Learning
Ensemble Learning
Transfer Learning
Generative Modeling
One-shot Learning
Active Learning
Semi-supervised Learning
Online Learning
Applications
Personalized Medicine
Cybersecurity and online Threat Detection, Anomaly Identification
Adaptive Tutoring
Image Analysis and Object Recognition
Conversational Agents and Language Translation
Self-driving Vehicles
Robotics
Learning Techniques
Online Learning
Imitation Learning
Data Extraction
Human-guided Training
Interactive Learning
Lifelong Learning
Reinforcement Learning

# Course Objectives

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- **Master ML Foundations**: Regression & Classification basics.
- **Understand Transformers**: Architecture and Attention mechanisms.
- **Hands on experience with RAG Systems**: Retrieval Augmented Generation implementation.
- **Agentic AI**: Deploy autonomous workflow agents.
- **AI Ethics**: Bias mitigation and fairness.

# Evaluation & Grading

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| Criterion | Date (expected) | Weightage (out of 50 marks) | Marks | Components |
|---|---|---|---|---|
| Attendance (over 85%) | NA | 5% (2.5 marks) | | |
| Contest 1 | 13 Feb | 20% (10 marks) | | |
| Contest 2 (case study based) | 27 March | 20% (10 marks) | | |
| Project Evaluation 1 | Feb 20 - Feb 28 | 25% (12.5 marks) | | Video, Mid sem Report, Presentation, Viva |
| Project Evaluation 2 | 5 March - 12 March | 30% (15 marks) | 50 | Video, End sem Report, Presentation |
| Mid Sem | 5 March - 14 March | | 20 | |
| End Sem | 6 May - 16 May | | 50 | |

# Phase 1: Foundations

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| Module | Key Topics |
|---|---|
| Intro to ML | AI vs ML vs Deep Learning, Bias-Variance Tradeoff |
| Preprocessing | Handling Missing Values, Feature Scaling, Encoding |
| Supervised Learning | Regression, Classification, Evaluation Metrics |
| Neural Networks | Perceptrons, Multi-layer Backpropagation, Optimizers |

# Phase 2: NLP & Architecture

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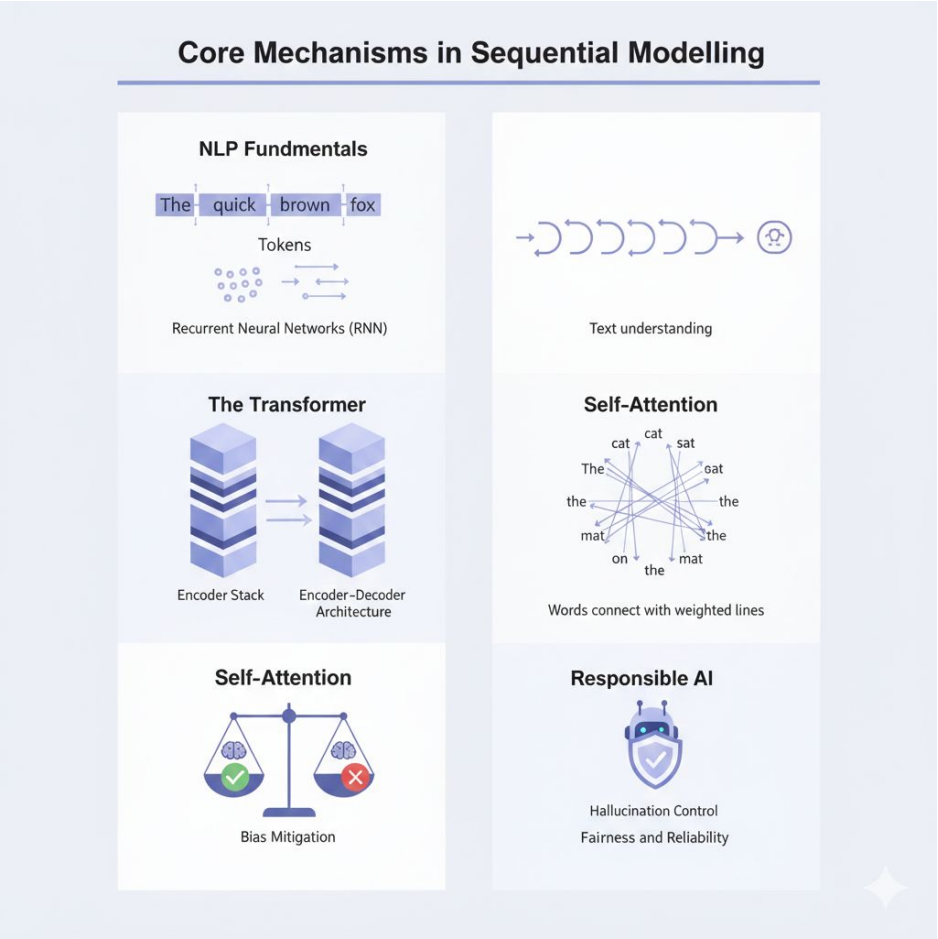
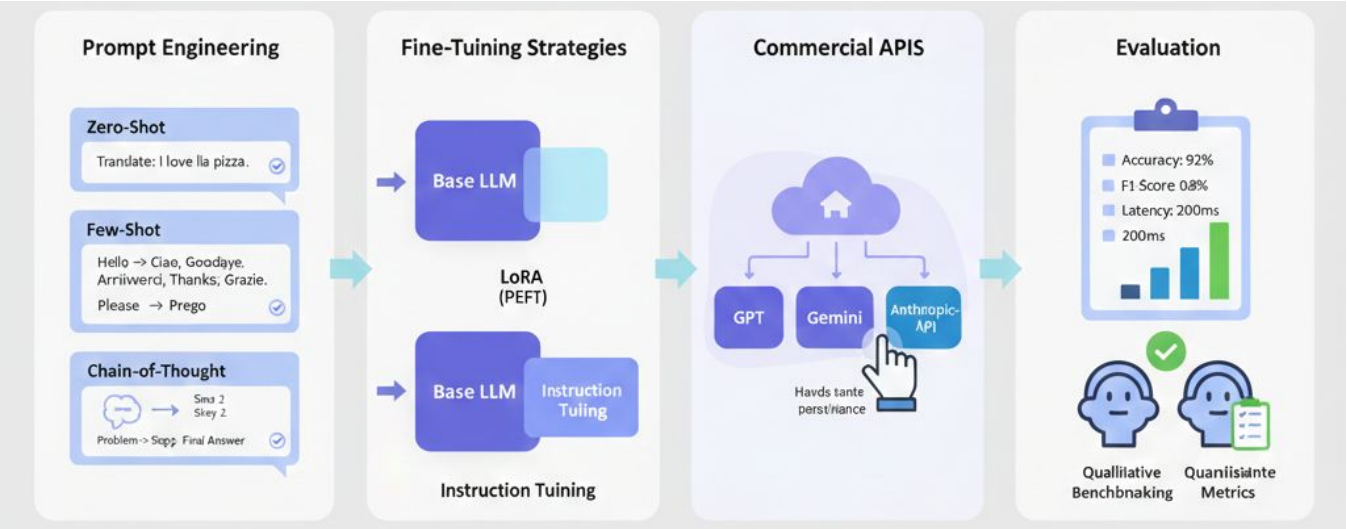
- **NLP Fundamentals:** Tokenization, Embeddings, RNNs.
- **The Transformer:** Encoder-Decoder structure.
- **Self-Attention:** Understanding the core engine.
- **Responsible AI: Mitigating Bias & Hallucination.**



# Phase 3: GenAI Engineering

- **Prompt Engineering:** Zero-shot, Few-shot, and Chain-of-Thought techniques.
- **Fine-Tuning Strategies:** Learning Instruction Tuning and parameter-efficient methods like LoRA (PEFT).
- **Commercial APIs:** Hands-on experience with OpenAI (GPT), Google (Gemini), and Anthropic models.
- **Evaluation:** Quantitative and qualitative benchmarking of LLM outputs.



# Phase 4: Advanced Systems

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- **RAG Architecture:** Building Retrieval Augmented Generation using Vector DBs like Chroma and FAISS.
- **Agentic AI:** Implementing autonomous agents using the ReAct framework and LangGraph.
- **Tool Integration:** Connecting LLMs to external calculators, search engines, and custom APIs.
- **Deployment:** Bringing models to production with Streamlit and cloud infrastructure.

# Common Questions

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1. If AI writes code, solves math, “thinks” philosophy why do I need to learn any of it?
2. If AI does writes code, would software development jobs vanish?
3. AI might take over humanity. Would it?

# Common Questions

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https://www.evidentlyai.com/blog/ai-failures-examples

# You are a VVIP

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Whom do you choose for your security?

# Your First AI Thought

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**One thing you want to build using AI by the end of this course.**

# Questions?